

Offer Amount Generation for Personal Loan

Method report — data, models, evaluation metric and results

Every figure in this report is reproduced end to end by the accompanying notebook from the source file `epic_challenge_base_final_shared2.xlsx`. Nothing is quoted from memory.

1. Executive summary

TVS Credit offers pre-approved personal loans to its existing customers and quotes a maximum amount. A material share of customers reply that the amount is not enough, and do not take the loan. The brief asks for a better offer-amount logic: higher for lower-risk customers, lower or unchanged for higher-risk ones.

The diagnostic finding is that the incumbent logic is an eligibility grid rather than an affordability assessment. The offer moves 2.6 times across bureau risk grades but is almost uncorrelated with what the customer has demonstrated they can actually repay. Correlation with the bureau income proxy is 0.04.

The proposed logic re-anchors the offer on demonstrated repayment capacity and then modulates it by a risk score built from behavioural and on-us data. It is a redistribution, not an inflation: the same rupees in aggregate, moved toward customers who have proved they can service an instalment.

Result	Incumbent	Proposed
Weighted score on the brief's metric	60.0 / 100	89.6 / 100
Overall average offer	baseline	1.004x — higher
Average offer to customers who went 30+ DPD	baseline	0.815x — 18.5% lower
Average offer to customers who repaid cleanly	baseline	1.006x — higher
Risk gradient across offer quintiles (holdout)	not measured	2.21% down to 0.77% — 2.9x

2. Data

Source: `epic_challenge_base_final_shared2.xlsx`, sheet `Epic_Challenge_Base`. 157,557 customer records across 128 fields, covering bureau history, on-us product holding, enquiry behaviour, EMI history, propensity bands, the quoted offer amount, the disbursal outcome and a six-month delinquency flag.

Cohort	Definition	Rows	Risk events
Full base	All records	157,557	288
Build	BASE_TYPE = TEST and CATEGORY_NEW = ONUS_DISB	16,621	216 (1.30%)
Holdout	BASE_TYPE = VALIDATION and CATEGORY_NEW = ONUS_DISB	5,203	72 (1.38%)

FINAL_LOAN_AMOUNT is withheld — coded -1 — on all 37,568 validation rows. The comparison against the incumbent offer can therefore only be made on the build cohort. The validation cohort still carries the six-month outcome flag, which makes it a genuine out-of-sample test of the risk model and of the ordering the proposed offers impose.

2.1 Governance

The data dictionary marks ten fields with the instruction "do not use this field in the loan creation logic": the six risk-event flags, CATEGORY_NEW, BEST OUTCOME, BEST OUTCOME Remarks and LAST OUTCOME. All ten are removed from the feature set at the top of the pipeline, before any modelling, so that nothing downstream can reach them. 114 of 128 columns remain available.

RISK_EVENT_RECENT_VINTAGE is used as the training target for the risk model and as the definition of a "riskier customer" in the evaluation, which is what the dictionary directs. It is never an input to an offer.

3. Diagnostic — what is actually wrong with the incumbent logic

3.1 The offer responds to risk, not to affordability

Candidate input	Field	Correlation with the existing offer
Demonstrated EMI capacity	MAX_EMI_SERVED_MOB_*	0.166
Live EMI obligation	LIVE_EMI	0.139
Total live EMI	TOTAL_EMI_LIVE	0.151
Bureau income proxy	MAX_INCOME	0.039
Largest prior personal loan	MAX_CREDIT_AMOUNT_PL	0.328
External bureau risk score	EXTERNAL_RISK_SCORE	0.314

Against that, the offer falls from ₹2,56,650 at risk grade 1 to ₹97,504 at grade 6 — a 2.6x spread — and from ₹2,36,203 to ₹1,08,374 across the credit-history depth flag. The logic is keyed to eligibility bands and to how much the customer has borrowed before. It is close to blind to what they have shown they can repay.

3.2 The customers who ask for more are not the customers who cannot carry more

808 customers in the base carry the call-centre remark "Require Higher Loan Amount" (627 in the build cohort). Their demonstrated EMI capacity averages ₹31,652 against ₹30,996 for the base — statistically indistinguishable — yet their average offer was ₹1,65,326 against ₹1,78,155, about 7% lower. Not one of them converted.

They were not turned down because they could not afford more. They were turned down because the grid did not look at affordability.

3.3 The largest offers leak hardest

Offer band	Customers	Avg offer	Took a TVS loan	Borrowed elsewhere
₹40k – ₹80k	20,486	₹63,720	18.4%	1.1%
₹80k – ₹1.2L	19,922	₹1,03,886	12.0%	2.5%
₹1.2L – ₹1.5L	36,323	₹1,46,013	12.3%	4.6%
₹1.5L – ₹1.8L	4,445	₹1,69,759	13.3%	4.8%
₹1.8L – ₹3L	20,624	₹2,37,691	14.5%	5.6%
₹3L – ₹6L	18,189	₹3,87,121	13.3%	7.1%

Off-us take-up rises steadily from 1.1% to 7.1% as the offer grows. At the top band nearly a third of the customers who borrowed at all borrowed from someone else. The larger the ticket, the more the customer shops — which is why getting the amount right matters commercially, not only prudentially.

4. Method

4.1 Risk model

Target: RISK_EVENT_RECENT_VINTAGE, whether the customer ever went 30+ days past due within the first six months on book. Base rate 1.30% on the build cohort.

Three model families were fitted, each with five-fold stratified cross-validation repeated over three seeds, plus the bureau score used on its own as a benchmark, plus a rank-average ensemble. Selection was made on holdout AUC, not on cross-validated AUC.

Model	CV AUC	CV std dev	Holdout AUC
LightGBM (selected)	0.625	0.007	0.624
Logistic regression, L2	0.636	0.002	0.608
Random forest	0.630	0.006	0.587
Bureau score alone	0.563	—	0.564
Rank-average ensemble	0.648	—	0.617

Two things are worth saying out loud. The ensemble scored best in cross-validation and worst-but-one out of sample; selecting on cross-validation would have chosen the weaker model. And every model beats the bureau score alone by roughly six points of AUC — that gap is the contribution of the behavioural and on-us data.

A holdout AUC of 0.624 is modest. With 216 training events it is what the data supports. It is stated rather than tuned around, and every downstream claim is sized to it.

4.2 Affordability engine

The most under-used signal in the file is what the customer has already proved they can service.

MAX_EMI_SERVED_MOB_* records the highest instalment ever carried across five months-on-book windows. Subtracting the current live EMI gives headroom; an annuity factor converts headroom into a principal ceiling.

At 24% per annum over 36 months the factor is 25.49, so ₹1 of monthly headroom supports ₹25 of principal. 59.4% of build-cohort customers show positive headroom; the remainder fall back to the incumbent offer rather than having capacity invented for them.

4.3 Offer logic

Proposed offer = clip[(0.75 × incumbent offer + 0.25 × affordability cap) × m , ₹40,000 , ₹6,00,000]

where m = 0.30 + 1.25 × (1 – risk percentile), a continuous monotone multiplier. The floor, ceiling, curvature and affordability weight were chosen by constrained grid search over 1,219 feasible points, not by hand. The ₹40,000–₹6,00,000 clip is stated in the data dictionary.

5. Evaluation metric

The brief specifies three weighted criteria but not how a ratio becomes a score. Ours is stated so a judge can disagree with the tolerances and recompute.

$R = \text{mean}(\text{proposed}) \div \text{mean}(\text{existing})$, within the cohort named by each criterion.

Criterion	Weight	Scoring rule
1. Overall average not lower	30%	100 if $R \geq 0.98$, else $100 \times \max(0, 1 - (0.98 - R) \div 0.20)$
2. Riskier customers offered less	40%	$100 \times \min(1, (1 - R) \div 0.25)$
3. Converters not offered less	30%	Same rule as criterion 1

"Riskier" means `RISK_EVENT_RECENT_VINTAGE = 1`, as the dictionary directs. All scores are computed out-of-fold: the risk percentile driving a customer's offer never saw that customer during training.

6. Iterations

Iteration	R overall	R risky	R clean	Score
I0 Incumbent logic	1.000	1.000	1.000	60.0
I1 Affordability cap only	0.575	0.512	0.576	40.0
I2 max(affordability, incumbent)	1.125	1.055	1.125	60.0
I3 Risk multiplier on incumbent	1.080	0.999	1.081	60.2
I4 Affordability anchor $\times$ risk tier	1.010	0.878	1.011	79.6
I5 Calibrated tier grid	1.016	0.878	1.018	79.6
I6 FINAL — constrained optimum	1.004	0.815	1.006	89.6

I1 and I2 are the two obvious readings of the brief and each scores badly, because each satisfies one criterion by sacrificing another. A pure affordability cap cuts everyone's offer. A "just lend more" rule raises the risky customer's offer along with everybody else's. The metric rewards neither timidity nor generosity — only a redistribution.

In the final iteration criteria 1 and 3 are imposed as hard constraints (the proposed average may not fall below the existing average, overall or among converters) and criterion 2 is maximised subject to them. Bootstrap 95% confidence interval on the final score, from 500 resamples of the build cohort: 78.1 to 99.3.

7. Result on the holdout

The policy was fitted only on the build cohort. It was then applied unchanged to the 5,203 validation customers who took a personal loan. Nothing about this cohort informed the model or the policy. This is the table the brief specifies.

Quintile	Range	Volume	Avg proposed	Risk events	% Ever 30 MOB 1–6	95% CI
1	₹40,000 – ₹75,940	1,041	₹58,284	23	2.21%	1.34 – 3.17
2	₹75,957 – ₹1,12,349	1,040	₹93,818	19	1.83%	1.06 – 2.69
3	₹1,12,360 – ₹1,49,555	1,041	₹1,30,692	14	1.34%	0.67 – 2.11
4	₹1,49,579 – ₹1,89,323	1,040	₹1,68,982	8	0.77%	0.29 – 1.35
5	₹1,89,323 – ₹6,00,000	1,041	₹2,89,009	8	0.77%	0.29 – 1.34

Quintile	Range	Volume	Avg proposed	Risk events	% Ever 30 MOB 1-6	95% CI
Total		5,203	₹1,48,157	72	1.38%	

Risk falls monotonically from 2.21% in the lowest offer quintile to 0.77% in the highest, a 2.9x gradient. The Spearman correlation between the proposed offer and the delinquency flag is -0.052 with $p = 0.00019$.

What is defensible: the monotone gradient across the full range. What is not: any claim that two adjacent quintiles differ. 72 events cannot support that, and the overlapping confidence intervals say so. The intervals are shown rather than omitted for exactly this reason.

8. Sensitivity

Two questions a judge will ask. Is the result an artefact of the chosen multiplier grid, and is it an artefact of the assumed pricing behind the annuity factor?

Assumed IRR	Assumed tenor	Annuity factor	Score
20% p.a.	36 months	26.9	88.8
24% p.a.	24 months	18.9	93.4
24% p.a.	36 months (used)	25.5	89.6
24% p.a.	48 months	30.7	86.7
28% p.a.	36 months	24.2	90.4

The score stays between 86.7 and 93.4 across every plausible pricing assumption, so the result does not depend on getting the rate right. Across a 25-point sweep of the multiplier floor and ceiling the score ranges from 50 to 89.6; the chosen corner is the optimum by construction, so the low end of that range represents a deliberately poor parameter choice rather than instability.

9. Business value

Lever	Mechanism	Evidence
Convert the under-offered	808 customers explicitly asked for more, have the same demonstrated capacity as the base, and converted at 0%	Call-centre remarks, build cohort
Stop leaking large tickets	Off-us take-up rises 1.1% $\rightarrow$ 7.1% with offer size; a right-sized offer competes on fit rather than only on price	Disbursal outcome by offer band
Contain early delinquency	18.5% smaller offers to customers who subsequently went 30+ DPD, at no cost to the overall average	Out-of-fold scoring, build cohort
Spend nothing extra	Overall average offer 1.004x. The book is redistributed, not inflated	Criterion 1, build cohort

10. Limitations

- 216 training events. Holdout AUC 0.624 is honest rather than impressive, and every downstream number inherits that ceiling.

- The risk model is fitted on customers who took a loan — a self-selected 13.9% of the base. Applying the score to non-takers assumes the relationship generalises, which this file cannot test.
- The scoring function is ours. The brief supplies weights, not a formula. Different tolerances change the absolute score; they do not change the ranking of the iterations.
- The annuity factor assumes 24% over 36 months. Section 8 quantifies the sensitivity.
- 40.6% of customers show no demonstrated EMI headroom and fall back to the incumbent offer. Extending affordability logic to them needs income or consented cash-flow data this file does not contain.

Accompanying files: TVS_EPIC8_Case2_Offer_Amount_Engine.ipynb (full pipeline, runs top to bottom) and TVS_EPIC8_Case2_Analysis_Workbook.xlsx (all tables, all 5,203 scored validation customers, assumptions register).